



MANUAL	
Document Number:	MAN-018-02
Effective Date:	13-AUG-2026
Version Number:	01
Page:	1 of 35

Title: Heidelberg SD-OCT Image Acquisition and File Export Manual - OCU410-301

Heidelberg SD-OCT Image Acquisition and File Export Manual

Protocol: OCU410-301



MANUAL	
Document Number:	MAN-018-02
Effective Date:	13-AUG-2026
Version Number:	01
Page:	2 of 35

Title: Heidelberg SD-OCT Image Acquisition and File Export Manual - OCU410-301

Contents

1. Study-Specific OCT Requirements	3
2. Instrumentation and Software Requirements	4
2.1 System Requirements	4
2.2 Alternate Heidelberg Spectralis Instrumentation and Software	4
2.3 Ability to Transfer Files Off of OCT System	4
3. Summary Procedures	5
4. Create a NORC Exports Folder on Your Desktop for Scan Exports	6
5. Check / Configure Heidelberg Spectralis Software	6
5.1 Check the Heidelberg SD-OCT Software Versions.....	6
5.2 Change the Destination Folder for Raw Data Exports to the “NORC Exports” Folder	9
6. Creating and locating a study patient in the Heidelberg SD-OCT system.....	11
6.1 Procedure for Creating the Study Patient Identity.....	11
7. SCAN ACQUISITION PROCEDURES	18
8. Export Procedures	23
9. NORC File Transmission Instructions.....	27
9.1 System Requirements	27
9.2 NORC File Transfer Accounts	27
9.3 Image Transmission Description.....	27
9.4 Using the NORC File Transfer Portal	28
9.5 Submission Follow-Up	28
10. Revision History.....	29
11. Signature Page.....	30
Appendix A	31
A. Purpose.....	31
B. Definitions and Grading Criteria.....	31



MANUAL	
Document Number:	MAN-018-02
Effective Date:	13-AUG-2026
Version Number:	01
Page:	3 of 35

Title: Heidelberg SD-OCT Image Acquisition and File Export Manual - OCU410-301

1. Study-Specific OCT Requirements

This section details the instrument-specific (i.e., Heidelberg Spectralis HRA + OCT) OCT requirements for the OCU410-301 Study.

Eyes to be Studied: Both eyes (OU)

Requirement	Macular Volume Scan	ONH Scan
Eyes	Both eyes (OU)	Both eyes (OU)
Imaging Mode	IR + OCT	—
Scan Pattern	Volume Scan (20° × 20°)	ONH Scan (15° × 15°)
B-Scan Sections	97	37
OCT ART Mean	9 frames (minimum)	9 frames (minimum)
Scan Quality	≥ 20	≥ 20
Field of View	30° Mode	30° Mode
OCT Volume Resolution	High Speed	High Speed

Required Images (OU):

- IR+OCT
- Separate IR (Field 2)
- Separate autofluorescence images (Field 2)

File Format and Submission:

Export Heidelberg E2E files via the NORC secure file transfer portal as the primary method for submitting imaging data, or CD/DVD/External drive if needed (see Section 10).

Required OCT Equipment and Software:

- Instrument: *Heidelberg Spectralis HRA + OCT
- HRA2 / Spectralis Family Acquisition Module Version: 4.0.0.0 or higher
- HRA / Spectralis Viewing Module Version: 4.0.0.0 or higher
- Heidelberg Eye Explorer (HEYEX) Version: 1.6.1.0 or higher

**See Section 2-B for alternate Heidelberg SD-OCT instrumentation and software.*



MANUAL	
Document Number:	MAN-018-02
Effective Date:	13-AUG-2026
Version Number:	01
Page:	4 of 35

Title: Heidelberg SD-OCT Image Acquisition and File Export Manual - OCU410-301

2. Instrumentation and Software Requirements

Please read this section carefully and contact NORC if you have any questions.

2.1 System Requirements

Acceptable instrumentation for this protocol is the Heidelberg Spectralis HRA + OCT, with HRA2 / Spectralis Family Acquisition Module Version 4.0.0.0 or higher, HRA / Spectralis Viewing Module Version 4.0.0.0 or higher, and Heidelberg Eye Explorer (HEYEX) Version 1.6.1.0 or higher.

Please note: This protocol has been written for the above-mentioned instrument and software versions. However, other instruments and software versions will also be accepted. These are listed in Section 2-B, below. Please keep in mind that some of the screen captures and procedure steps presented in this manual may be different for the different instruments listed in Section 2-B.

2.2 Alternate Heidelberg Spectralis Instrumentation and Software

Instrument: Heidelberg Spectralis OCT

Software Version: 5.0 or higher; Spectralis Family Acquisition Module Version: 5.0.2.0 or higher; Heidelberg Eye Explorer (HEYEX) Version: 1.6.1.0 or higher

Instrument: Heidelberg Spectralis OCT Plus

Software Version: 4.0 or higher; Spectralis Family Acquisition Module Version: 4.0.0.0 or higher; Heidelberg Eye Explorer (HEYEX) Version: 1.6.1.0 or higher

Instrument: Heidelberg Spectralis FA + OCT

Software Version: 4.0 or higher; Spectralis Family Acquisition Module Version: 4.0.0.0 or higher; Heidelberg Eye Explorer (HEYEX) Version: 1.6.1.0 or higher

(If you do not have updated software, please contact your Heidelberg Spectralis representative to request a software upgrade.)

2.3 Ability to Transfer Files Off of OCT System

Your system must have the capability to transfer digital files off the Heidelberg SD-OCT computer so that they can be transmitted to NORC via the online portal. The easiest way to do this may be to connect a USB device such as a flash drive to the Heidelberg SD-OCT machine, the contents of which can be sent directly to NORC or transferred to another computer to be uploaded via web-transmission to NORC.

Please be sure that you have the above hardware and software capabilities before beginning study activities.



MANUAL	
Document Number:	MAN-018-02
Effective Date:	13-AUG-2026
Version Number:	01
Page:	5 of 35

Title: Heidelberg SD-OCT Image Acquisition and File Export Manual - OCU410-301

3. Summary Procedures

This section summarizes the study-specific SD-OCT procedures for the actual patient visit. Images for qualification should also follow these procedures.

- Create an export location (Section 4) – Note: this only needs to be done once.
- Check / configure Heidelberg SD-OCT software (Section 5)
- Check / configure Windows Explorer (Section 6)
- Create, or locate the study patient in the Heidelberg SD-OCT database (Section 7). If available, USE HEYEX FOLLOW-UP FUNCTION
- See Section 8 for Scan Acquisition settings.
- Scan the Patient (Section 8).
- Verify that scan quality is ≥ 20 . The Scan Quality factor will be displayed on the screen. Rescan if needed to acquire a good quality scan (Section 8).
- Save the scan (Section 8).
- Export the data files for a single patient visit to the “NORC Exports” folder (Section 9).
- Upload via the NORC secure file transfer portal (Section 10), or CD/DVD/USB/HARD DRIVE.

Scans to Acquire

For both site qualification and study subject imaging, acquire the required Macular Volume and ONH scans for both eyes (OU) using the study-specific scan parameters listed in **Section 1**.



MANUAL	
Document Number:	MAN-018-02
Effective Date:	13-AUG-2026
Version Number:	01
Page:	6 of 35

Title: Heidelberg SD-OCT Image Acquisition and File Export Manual - OCU410-301

4. Create a NORC Exports Folder on Your Desktop for Scan Exports

This section gives procedures for creating a folder on your desktop into which you will export the NORC study scans. This procedure only needs to be done once to set up your system.

Start from your Windows Desktop. Right-click on the desktop and select New Folder. When the new folder icon appears, type “NORC Exports (Figure 1).”

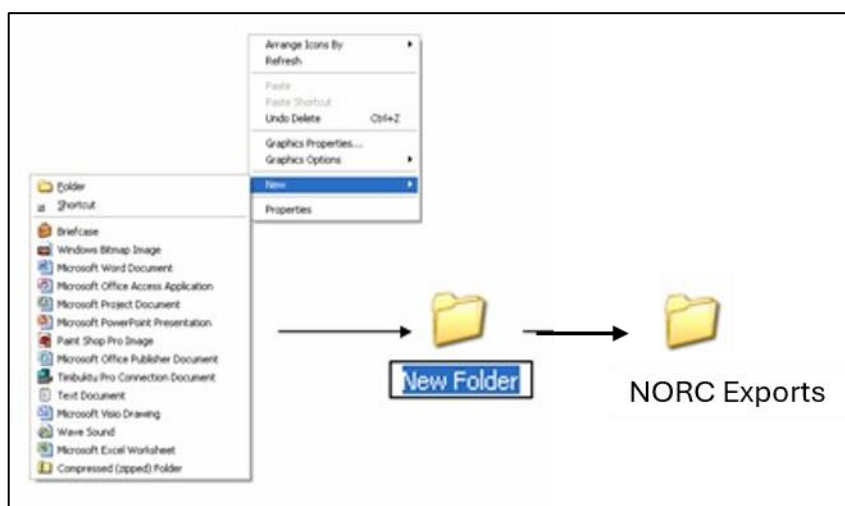


Figure 1. Create a “NORC Exports” folder on the Desktop.

5. Check / Configure Heidelberg Spectralis Software

This section explains how to:

- Ensure that the Heidelberg Eye Explorer (HEYEX) Version is 1.6.1.0 or higher.
- Ensure that the HRA2 / Spectralis Family Acquisition Module Version is 4.0.0.0 or higher.
- Ensure that the HRA / Spectralis Viewing Module Version is 4.0.0.0 or higher.
- Change the destination folder for Raw Data Exports to the “NORC Exports” folder.

Note: See Section 2.2 for alternate hardware and software versions.

5.1 Check the Heidelberg SD-OCT Software Versions

1. Double-click on the Heidelberg Eye Explorer Icon.
2. When the Main Screen opens, select Help - About from the Main Menu (Figure 2).



MANUAL	
Document Number:	MAN-018-02
Effective Date:	13-AUG-2026
Version Number:	01
Page:	7 of 35

Title: Heidelberg SD-OCT Image Acquisition and File Export Manual - OCU410-301

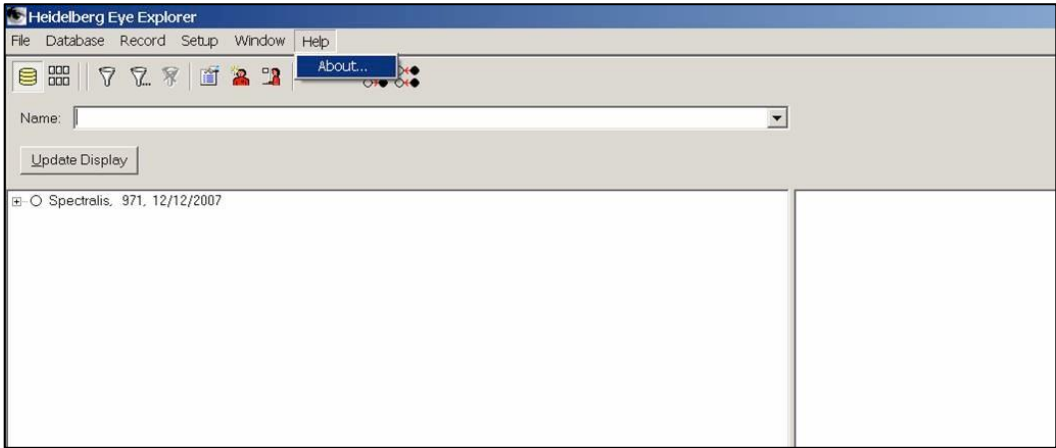


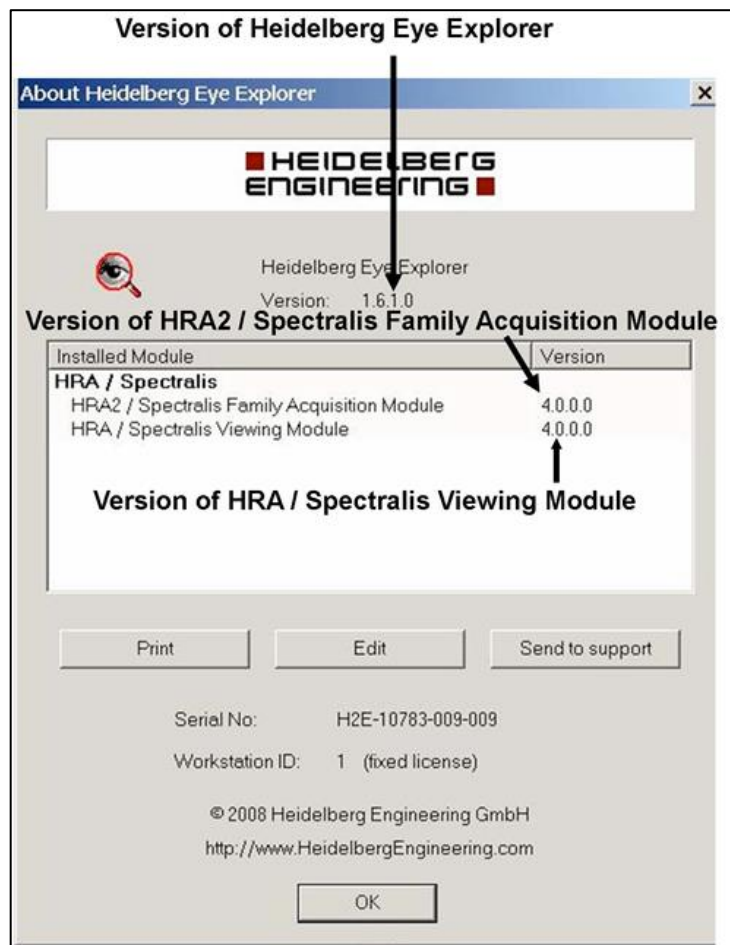
Figure 2. Access “About Heidelberg Eye Explorer” Information.

3. The “About Heidelberg Eye Explorer” box will be displayed, showing the Version of Heidelberg Eye Explorer, the Version of HRA2 / Spectralis Family Acquisition Module, and the Version of HRA / Spectralis Viewing Module (Figure 3).



MANUAL	
Document Number:	MAN-018-02
Effective Date:	13-AUG-2026
Version Number:	01
Page:	8 of 35

Title: Heidelberg SD-OCT Image Acquisition and File Export Manual - OCU410-301



*Figure 3. Software Versions.*4. If the version of Heidelberg Eye Explorer is not 1.6.1.0 or higher, the version of HRA2 / Spectralis Family Acquisition Module is not 4.0.0.0 or higher and/or the version of HRA / Spectralis Viewing Module is not 4.0.0.0 or higher, please contact your Heidelberg representative for a software upgrade.



MANUAL	
Document Number:	MAN-018-02
Effective Date:	13-AUG-2026
Version Number:	01
Page:	9 of 35

Title: Heidelberg SD-OCT Image Acquisition and File Export Manual - OCU410-301

5.2 Change the Destination Folder for Raw Data Exports to the “NORC Exports” Folder

5.2.1 Launch the Heidelberg Eye Explorer software by double-clicking on the software icon.

5.2.2 When the Main Screen opens, click on Setup - Options from the Main Menu (see Figure 4).

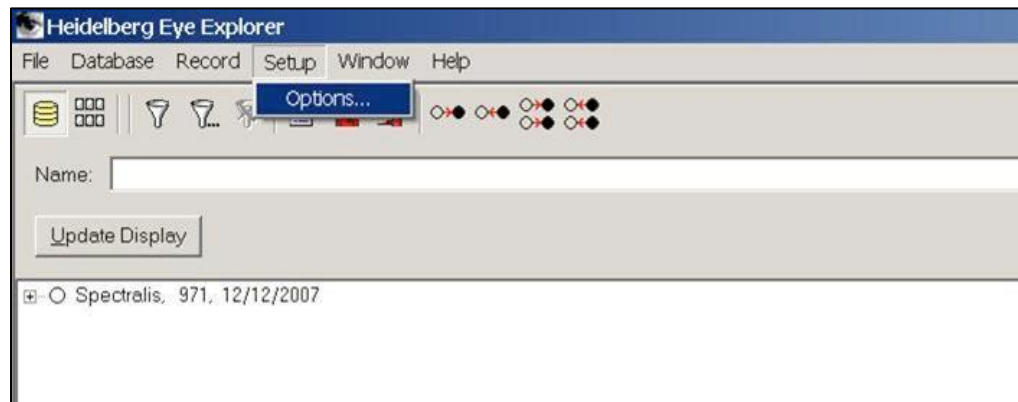


Figure 4. Setup “NORC Exports” Folder for Scan Exports

5.2.3 When the “Options” box opens, click the “Plug Ins” tab, select “hraviewer” in the selection window, and click Setup (see Figure 5).



MANUAL

Document Number:	MAN-018-02
Effective Date:	13-AUG-2026
Version Number:	01
Page:	10 of 35

Title: Heidelberg SD-OCT Image Acquisition and File Export Manual - OCU410-301

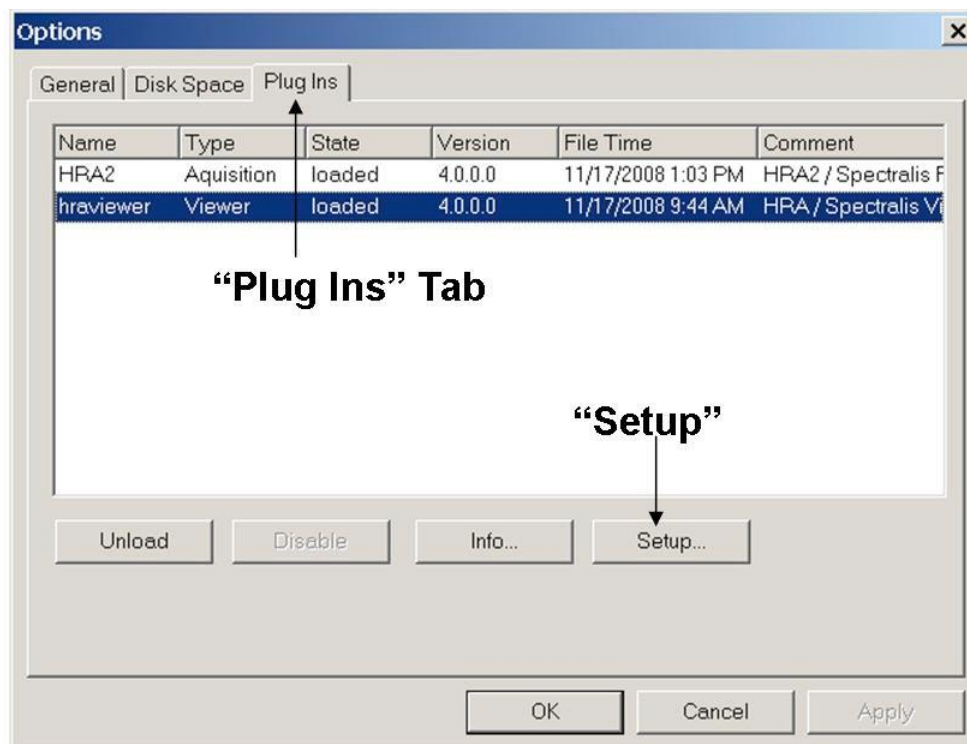
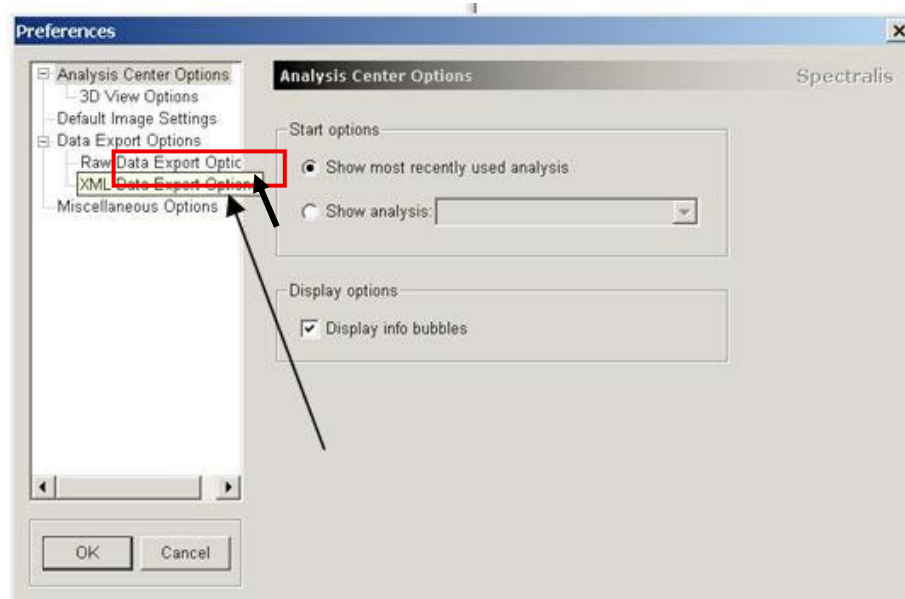


Figure 5. “hra viewer” Highlighted in Selection Window on “Plug Ins” Tab

5.2.4 When the “Preferences” box opens, click “Raw Data Export Options” in the “Preferences” (selection) window (see Figure 6). The Raw Data Export Options view will be displayed.





MANUAL	
Document Number:	MAN-018-02
Effective Date:	13-AUG-2026
Version Number:	01
Page:	11 of 35

Title: Heidelberg SD-OCT Image Acquisition and File Export Manual - OCU410-301

Figure 6. “Raw Data Export Options”

5.2.5 On the “Preferences” box, browse to the “NORC Exports” destination folder in the “Destination folder for export files” section, leave the “post-processing application” field blank, and select Graphics file format as E2E (Figure 7).

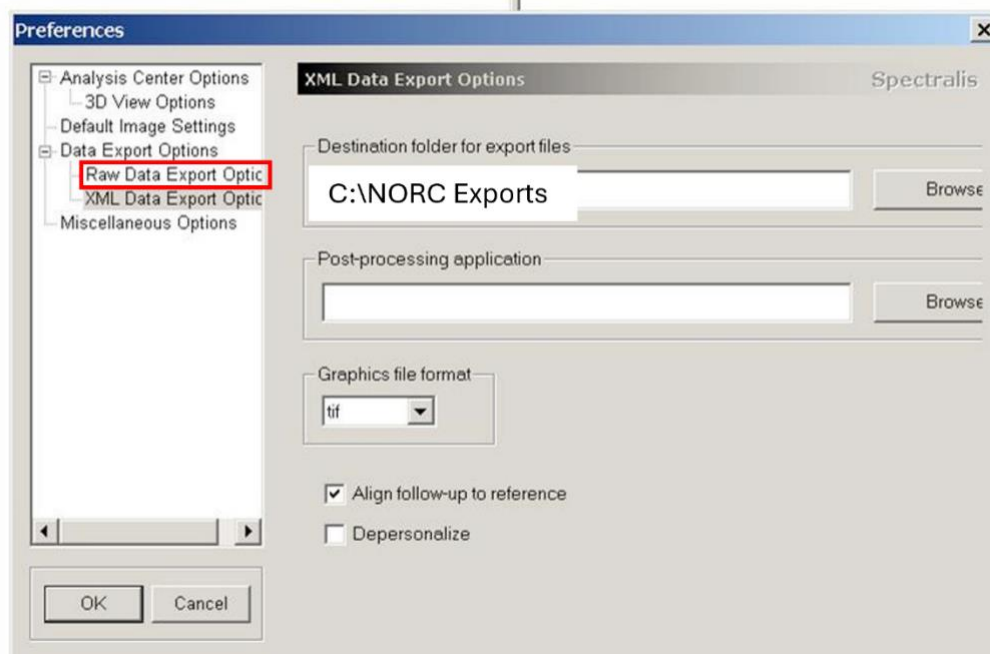


Figure 7. “Preferences” Box

6. Creating and locating a study patient in the Heidelberg SD-OCT system

These procedures describe how to create a new study patient in the Spectralis HRA + OCT system, and to locate a study patient that has already been created.

The first time you perform imaging on a new study patient, you will create the study patient using the identity masking procedures provided in this section.

6.1 Procedure for Creating the Study Patient Identity

6.1.1 Launch the Spectralis HRA + OCT software program by double-clicking on the



Heidelberg Eye Explorer icon. The “Main Screen / Database Window” will

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MANUAL	
Document Number:	MAN-018-02
Effective Date:	13-AUG-2026
Version Number:	01
Page:	12 of 35

Title: Heidelberg SD-OCT Image Acquisition and File Export Manual - OCU410-301

open (see Figure 8). See Figure 9 for shortcut buttons.

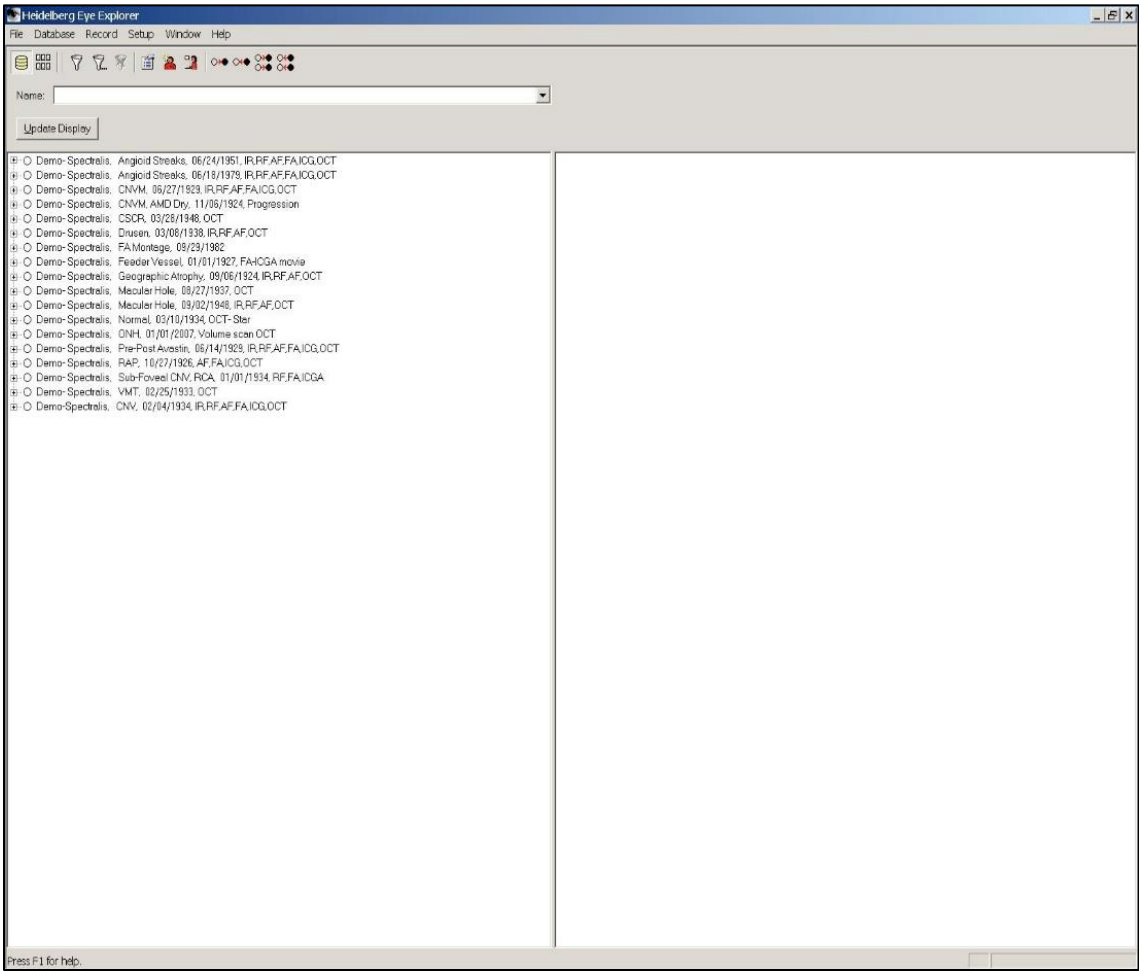


Figure 8. Main Screen / Database Window



MANUAL	
Document Number:	MAN-018-02
Effective Date:	13-AUG-2026
Version Number:	01
Page:	13 of 35

Title: Heidelberg SD-OCT Image Acquisition and File Export Manual - OCU410-301



Figure 9. Shortcut buttons

6.1.2 On the “Main Screen / Database Window”, click on the “New Patient” icon (Figure 10).



Figure 10. New Patient Icon

6.1.3 When the “Patient Data” box opens, enter the study patient’s data as listed below, depending on whether you are submitting a study patient visit (Figure 11), or qualification images.



MANUAL	
Document Number:	MAN-018-02
Effective Date:	13-AUG-2026
Version Number:	01
Page:	14 of 35

Title: Heidelberg SD-OCT Image Acquisition and File Export Manual - OCU410-301

For study patients-

Complete the patient demographic fields according to the Sponsor Guidelines. Unless otherwise instructed by the sponsor, leave all remaining fields blank.

The screenshot shows a 'Patient Data' dialog box with the following fields and values:

- Patient-DB-ID: 0
- Last name: SPONSOR AND PROTOCOL
- First name: SUBJECT ID NUMBER
- Title: (empty)
- Date of birth: 1/1/1900
- Sex: ☒ Female ☐ Male
- Patient-ID: (empty)

Buttons: Import, OK, Cancel, Apply.

Figure 11. Patient Data Box

6.1.4 Click **OK** (see Figure 11. The “Examination Data” box will open (Figure 12).



MANUAL	
Document Number:	MAN-018-02
Effective Date:	13-AUG-2026
Version Number:	01
Page:	15 of 35

Title: Heidelberg SD-OCT Image Acquisition and File Export Manual - OCU410-301

The **Examination Data** dialog box has two tabs: **Examination Data** and **Diagnosis**. The **Examination Data** tab is active. It contains the following fields:

- Patient:** SPONSOR AND PROTOCOL, SUBJECT ID NUMBER.
- Date/Time:** 08/21/2009
- Device type:** Spectralis HRA + OCT (dropdown menu)
- Operator:** BB (dropdown menu)
- Study:** (dropdown menu) and a button with three dots (...)

At the bottom right are three buttons: **OK**, **Cancel**, and **Apply**.

Figure 12. “Examination Data” box

6.1.5 On the *Examination Data* box, select “Spectralis HRA + OCT” from the “Device type” drop-down menu (if you are using an instrument other than the Spectralis HRA + OCT, select the appropriate device type, corresponding to your specific instrument). Select your initials from the “Operator” drop-down menu (if your initials are not available in the drop-down menu, type in your initials, and it will get saved in the drop-down menu automatically). Leave the “Study” field blank. Click **OK**. See Figure 12, above.

6.1.6 The “Eye Data” Box will open (Figure 13).

The **Eye Data** dialog box is divided into two columns: **OD** and **OS**. Each column contains the following fields:

- C-Curve [mm]:** 7.7
- Refraction [dpt]:** 0
- Cylinder [dpt]:** 0
- Axis [deg]:** 0
- Pupil size [mm]:** 0
- IOP [mmHg]:** 0
- VFieldMean:** 0
- VFieldVar:** 0
- Corrective Lens:** None (dropdown menu)

At the bottom are two buttons: **OK** and **Cancel**.

Figure 13. “Eye Data” box



MANUAL	
Document Number:	MAN-018-02
Effective Date:	13-AUG-2026
Version Number:	01
Page:	16 of 35

Title: Heidelberg SD-OCT Image Acquisition and File Export Manual - OCU410-301

- 6.1.7 **DO NOT** enter/change any data in the “Eye Data” Box. (Leave all default settings in all the fields, as is / unchanged.) See Figure 13, above. Click **OK**
- 6.1.8 A preliminary Image Acquisition window will be displayed (Figure 14). If you are ready to scan the patient for the Baseline visit, go ahead to section 6B. (Now that you have set up the patient in the database, you also have the option of scanning the patient at a later time by calling up the patient by performing a patient-search.)

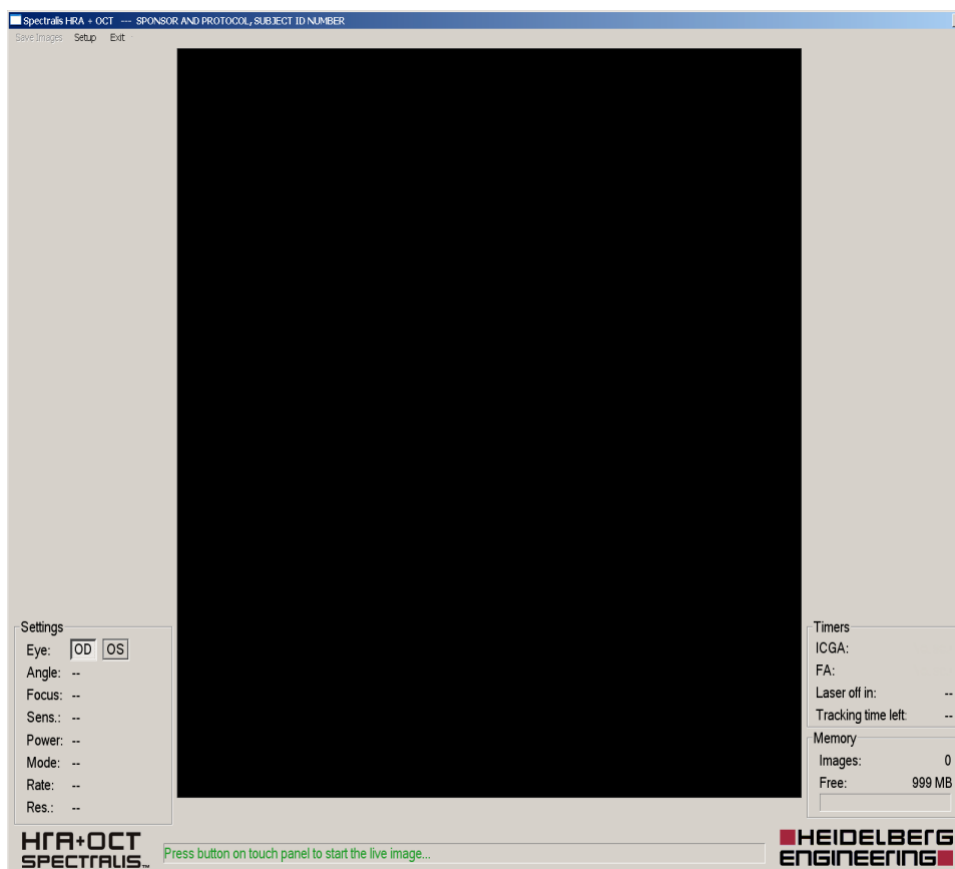


Figure 14. Preliminary Image Acquisition Screen

6.2 Locate the Patient for Scanning

You should have already set up the patient in the Heidelberg SD OCT Database

- 6.2.1 From the main screen, perform a patient-search by typing the sponsor (and protocol, if needed) in the “Name” field and clicking **Update Display** (see Figure 15)



MANUAL	
Document Number:	MAN-018-02
Effective Date:	13-AUG-2026
Version Number:	01
Page:	17 of 35

Title: Heidelberg SD-OCT Image Acquisition and File Export Manual - OCU410-301

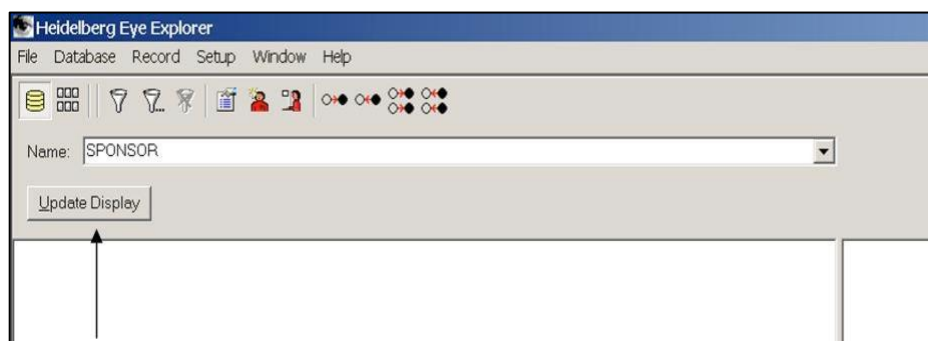


Figure 15. Patient-search

6.2.2 When the study patient is displayed, select the patient and click on the “New Examination” icon (see Figure 16).

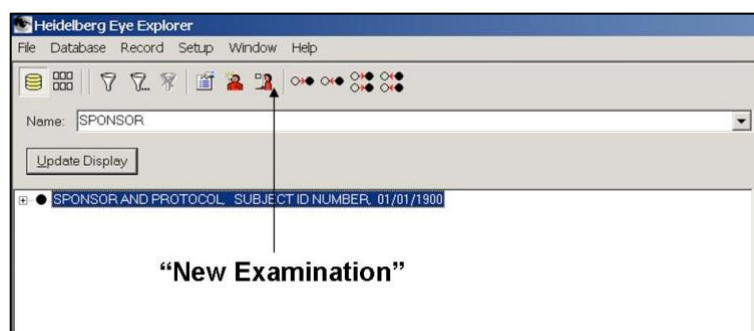


Figure 16. Highlight Patient and click “New Examination”

6.2.3 When the “Continue Examination” Box opens, click **Yes** (See Figure 17).

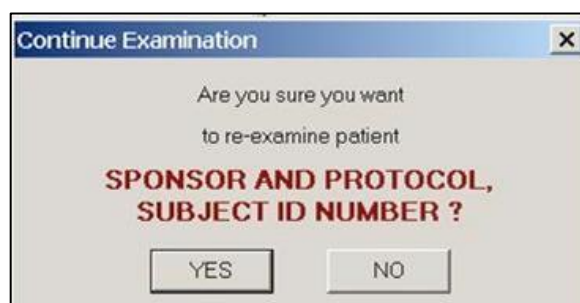


Figure 17. Confirm Patient Re-examination



MANUAL	
Document Number:	MAN-018-02
Effective Date:	13-AUG-2026
Version Number:	01
Page:	18 of 35

Title: Heidelberg SD-OCT Image Acquisition and File Export Manual - OCU410-301

7. SCAN ACQUISITION PROCEDURES

- After the preliminary Image Acquisition window opens (Figure 14), the Laser On/ Off button on the Control Panel will automatically turn from red to yellow.
- Perform the following steps on the Control panel:

(Note: Control Panel display colors; inactive / unselected buttons will display in Red, and active / selected buttons will display in blue.)

- Press the Yellow On / Off button on the Control Panel to activate the laser and camera. The screen will change as shown in Figure 22, and the Acquisition Menu will be displayed on the Control Panel (see Figure 19).

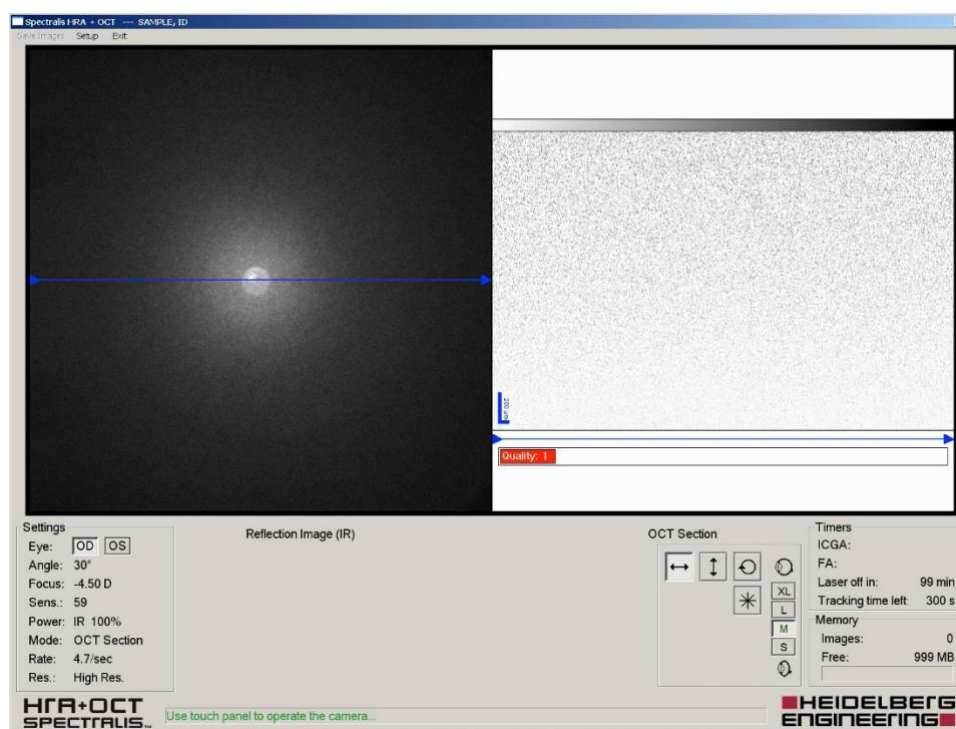


Figure 18. Screen after Laser and Camera are Activated



MANUAL	
Document Number:	MAN-018-02
Effective Date:	13-AUG-2026
Version Number:	01
Page:	19 of 35

Title: Heidelberg SD-OCT Image Acquisition and File Export Manual - OCU410-301

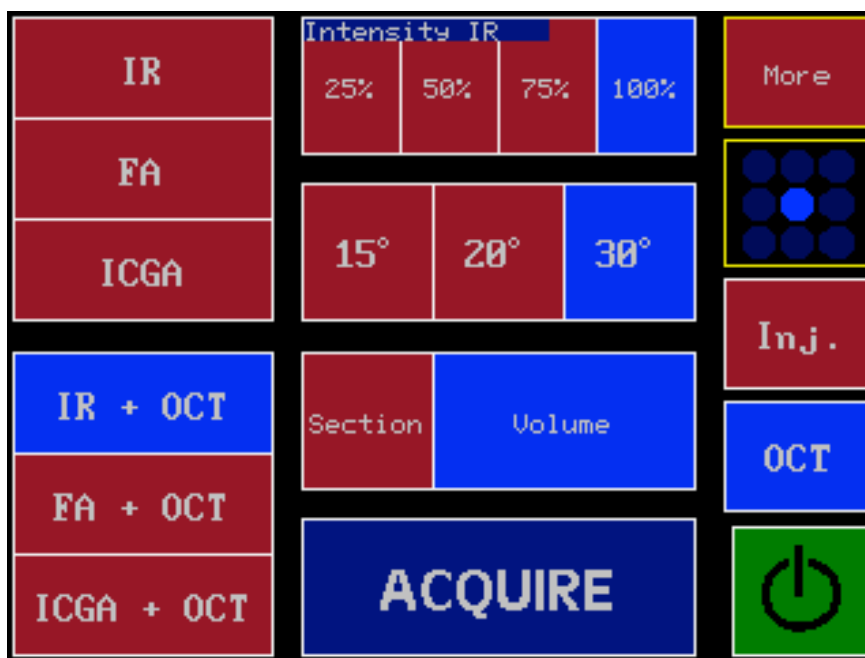


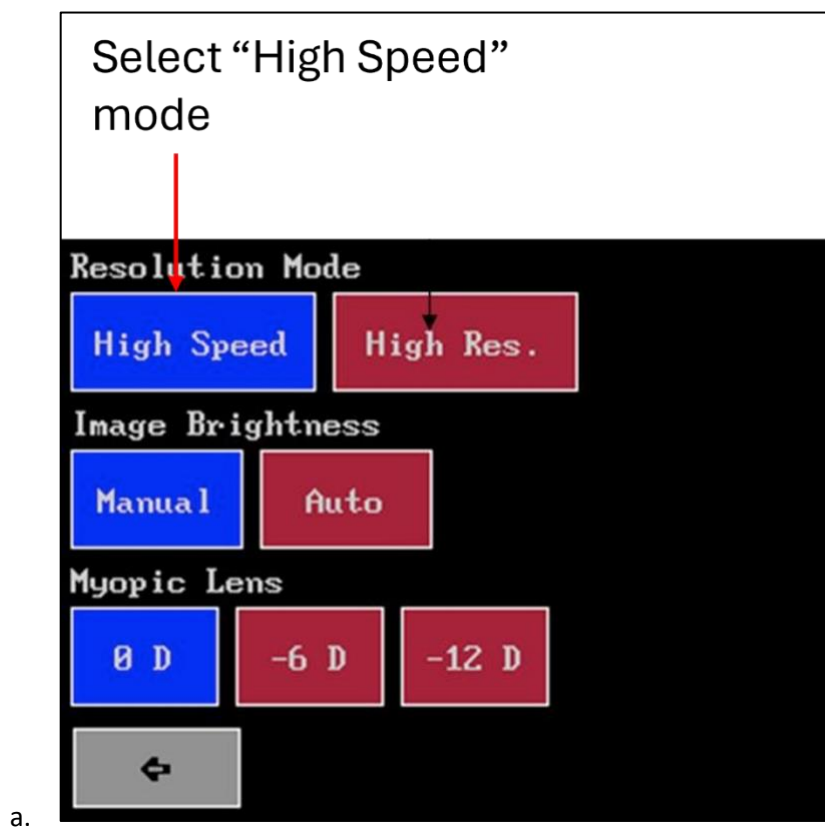
Figure 19. Control Panel

- ii) Make sure that the “OCT” button is selected (displayed in blue). If not, select it. See Figure 19.
- iii) Select the IR + OCT button (display it in blue). See Figure19.
- iv) Make sure that the “Volume” button is selected (displayed in blue). If not, select it. See Figure 19.
- v) Make sure that the Field of View is 30° (displayed in blue). If not, select it. See Figure 19.
- vi) Select the appropriate IR laser intensity to adjust for patient media (Typically, 50 – 75%). The default setting is 100%. See Figure 19.
- vii) If the Resolution setting on your Control Panel does not default to “High Res.”, select the “More” button on the Acquisition Menu, and when the Resolution Mode menu is displayed, select “High Speed.” (see Figure 20). Click on the back arrow to go back to the Acquisition menu.



MANUAL	
Document Number:	MAN-018-02
Effective Date:	13-AUG-2026
Version Number:	01
Page:	20 of 35

Title: Heidelberg SD-OCT Image Acquisition and File Export Manual - OCU410-301



b. Figure 20. Resolution Mode – High Speed.

- viii) Press the Sensitivity / ART button located below the touch pad of the Control Panel (see Figure 21). The screen will change as shown in Figure 22, with the addition of a small ART (Automatic Real Time) image window at the bottom of the screen.

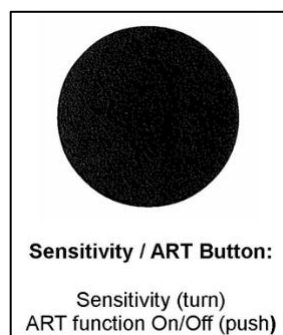


Figure 21. Sensitivity / ART Button



MANUAL	
Document Number:	MAN-018-02
Effective Date:	13-AUG-2026
Version Number:	01
Page:	21 of 35

Title: Heidelberg SD-OCT Image Acquisition and File Export Manual - OCU410-301

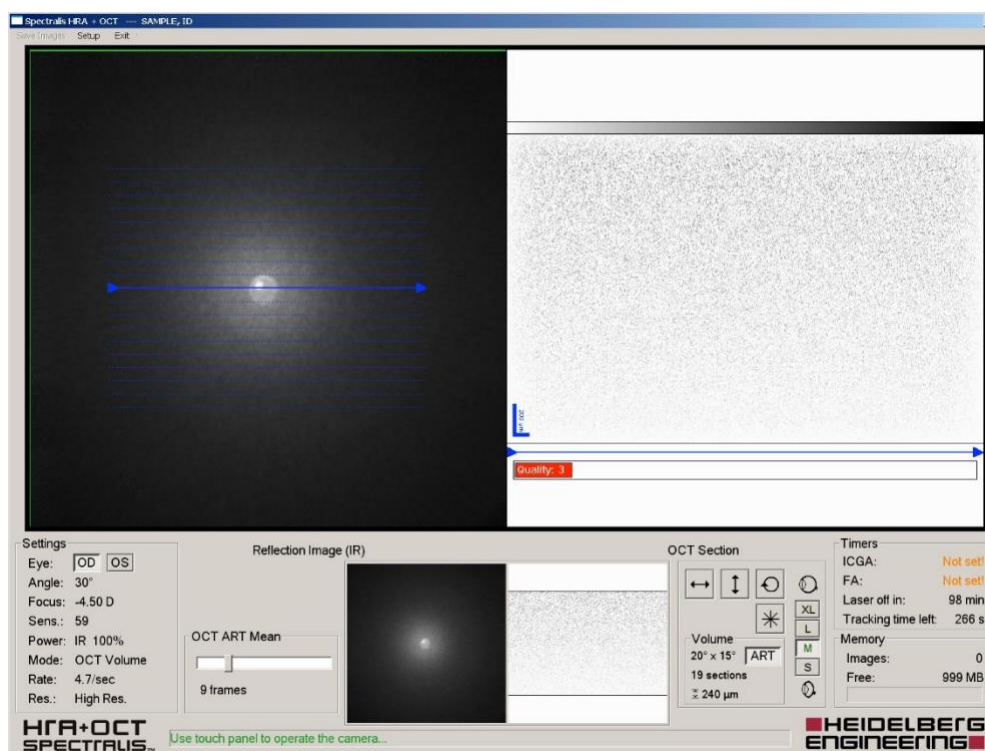


Figure 22. Screen with a small ART Image Window at bottom of Screen

- Bring the camera and patient into position, adjust alignment, distance, and focus to optimize image quality.
- Utilize the OCT Acquisition window on the monitor to align the camera with the Fundus image on the left side of the screen. A cross-sectional image of the retina will appear in the OCT image window (see Figure 23).
- Focus the IR photograph on the blood vessels and ensure that the OCT image on the right side of the screen is no lower than in the center of the OCT image window
- Select / check the following settings on the OCT Acquisition window:
 - Scan Pattern:** check that **Volume Scan (20° x 20°)** is displayed.
 - Number of Sections in the scan pattern:** the number of sections defaults to 19. Increase the sections to **97** by pressing the Shift key + the Up (↑) Arrow key on the keyboard until 97 sections is reached. If it is too difficult to acquire these many scans because the patient is poor fixator and scan time is too long, it is okay to reduce to 49 B-scans (please inform the situation to NORC).
 - OCT ART Mean:** ensure that an OCT ART Mean of **15 frames** is selected and



MANUAL	
Document Number:	MAN-018-02
Effective Date:	13-AUG-2026
Version Number:	01
Page:	22 of 35

Title: Heidelberg SD-OCT Image Acquisition and File Export Manual - OCU410-301

displayed (to increase or decrease the number of frames, move the bar under OCT ART Mean to the right or left respectively). If you have difficulty with capturing at an ART of 15 (e.g., patient is a poor fixator and scan time is too long, it is okay to reduce the ART, though ART levels below 9 may reduce the quality of our grading.

- iv) **Scan Quality:** ensure that a Scan Quality Factor of ≥ 20 is reached.
- v) **Eye Length:** select “M” (medium). If focus cannot be obtained at this setting, adjust the default Eye Length of Medium to “S” (short), “L” (long), or “XL” (extra-long) until the retina is focused.
- vi) **Eye (OD/OS):** the Eye will automatically be selected depending on the position of the camera. Therefore, no selection is required as to OD or OS.
- vii) **Field of View:** check that the Angle is at 30° (this selection was made on the Control Panel).
- viii) **Mode:** Check that Mode is **OCT Volume**.
- ix) **Resolution:** check that “Res.” Is “High Speed.”.
- When all the required scan parameters are selected and/or optimized, press **ACQUIRE** on the Control Panel.
- After the scan is complete, if the visit requires both eyes to be scanned, move the camera backwards and over, to focus on the opposite eye, and acquire the scan of the opposite eye by following Steps 3 – 7 above. *(Note: You will be aware that the scan is complete, when the small OCT ART Image window disappears from the bottom of the screen, and the B-scans have migrated from the bottom to the top of the IR image).*
- Click “Save” on the upper left side of the screen (see Figure 23) on the Menu Bar to save the OCT scan/s. *NOTE: Scans of both eyes can be saved at the same time.*
- Click “Exit” on the upper left side of the screen on the Menu Bar to exit the OCT Image Acquisition screen.
- Proceed to the next section (8) for export procedures.



MANUAL	
Document Number:	MAN-018-02
Effective Date:	13-AUG-2026
Version Number:	01
Page:	23 of 35

Title: Heidelberg SD-OCT Image Acquisition and File Export Manual - OCU410-301

8. Export Procedures

Export all Spectralis data (OCT, IR and FAF images for each eye or each subject) in .E2E format.

1. To begin exporting the scan in .E2E format, right-click on the icon representing the scan that needs to be exported. The options list-box with the Export functionality enabled will be displayed. Hover over "Export," and a subsidiary options list-box with export file formats will be displayed. Click on an E2E, then repeat steps to complete the export.
2. After the export of .E2E has been completed, using Windows Explorer, locate and double-click on the "NORC Exports" folder to open it. From the Windows menu, select View - Thumbnails. You should see, for a single eye, one .E2E file for each eye or each imaging scan (all images, macular volume scan, ONH scan, IR images and FAF images of each subject (2 eyes) can be exported in a single .E2E file).

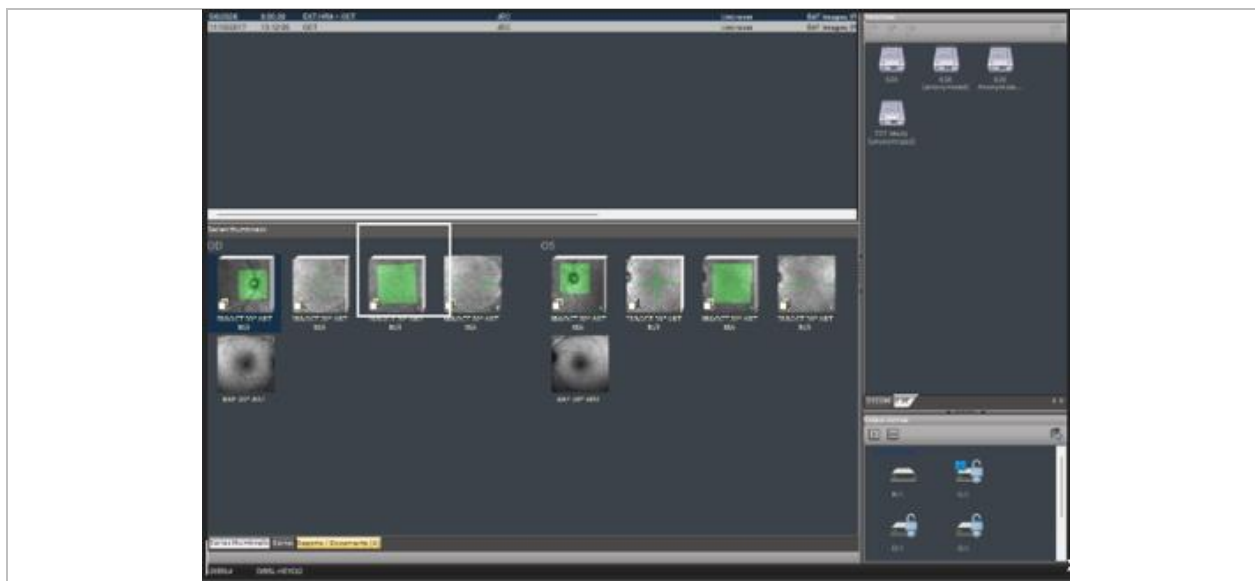
HRA2 OCT Volume Data Exporting Procedure

Before You Begin

- ☐ Confirm the subject's OCT volume(s) have been fully acquired and saved in HEYEX before starting the export process.
- ☐ Have the subject's Subject ID and Visit Number available for the naming step.
- ☐ Confirm the study-specific data folder has already been created on the desktop.

Step 1 Select the OCT Volume

Select the OCT volume(s) to be exported, as indicated by the white box in the HEYEX interface.



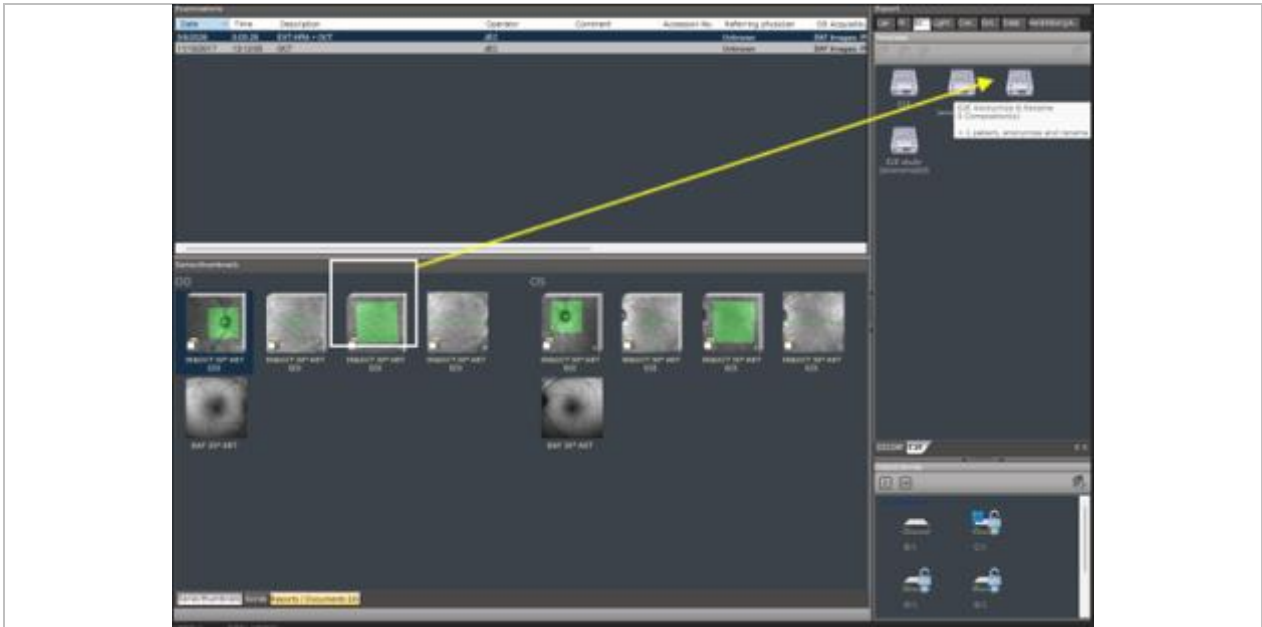


MANUAL	
Document Number:	MAN-018-02
Effective Date:	13-AUG-2026
Version Number:	01
Page:	24 of 35

Title: Heidelberg SD-OCT Image Acquisition and File Export Manual - OCU410-301

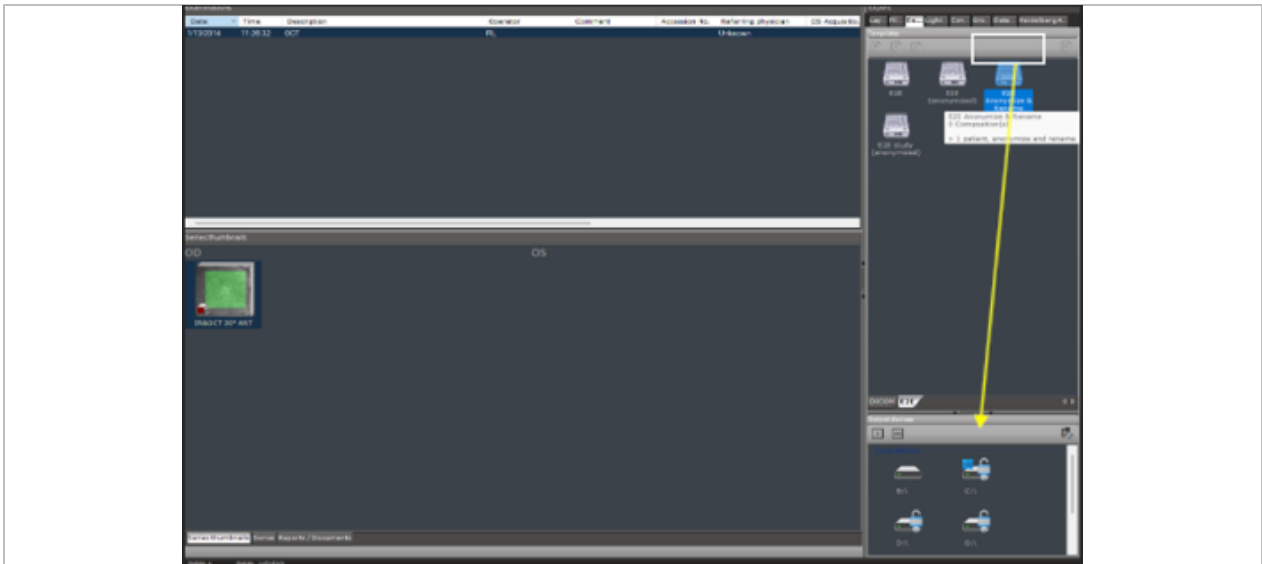
Step 2 Export the OCT Volume

In the **Export** tab on the right side of the HEYEX window, drag and drop the selected OCT volume(s) into the "E2E Anonymized & Rename Drive."



Step 3 Rename the Exported Data

Double-click "E2E Anonymized & Rename Drive." This will allow you to rename the exported E2E volume according to the study-specific naming instructions.





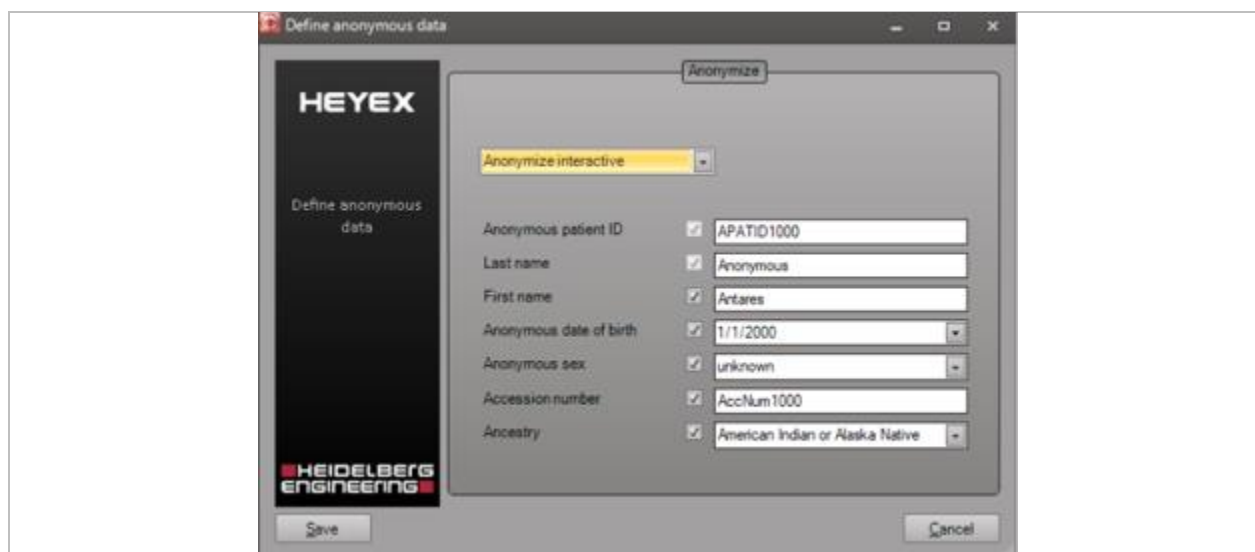
MANUAL

Document Number:	MAN-018-02
Effective Date:	13-AUG-2026
Version Number:	01
Page:	25 of 35

Title: Heidelberg SD-OCT Image Acquisition and File Export Manual - OCU410-301

Step 4 Apply the Study-Specific Naming Convention

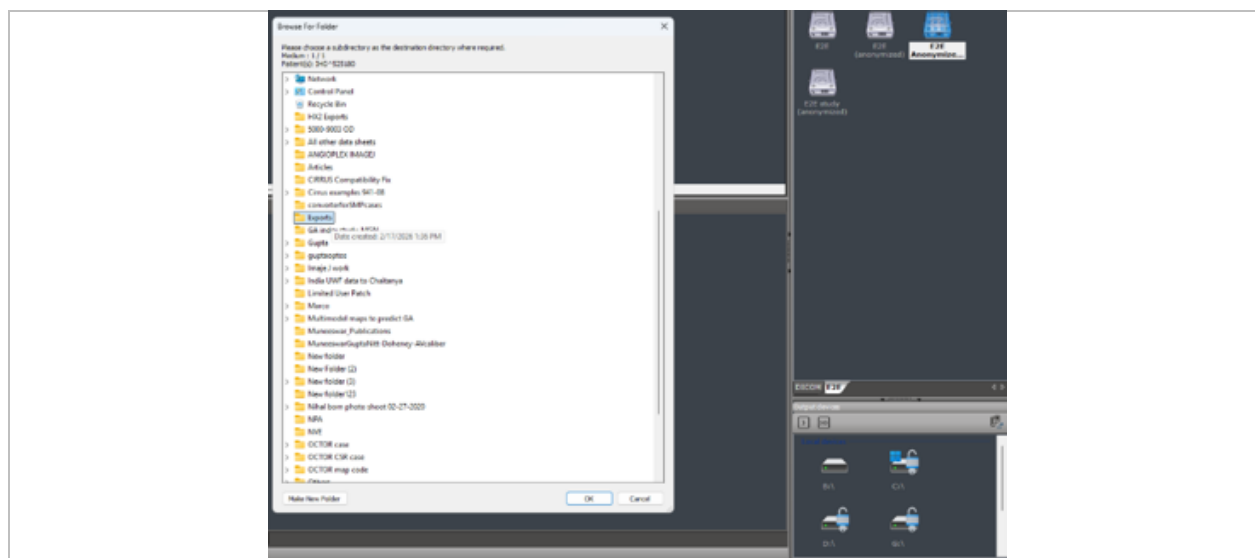
The photographer should rename the exported E2E volume according to the study naming convention recommended by the sponsor.



Step 5 Transfer the Renamed Data to the Study Folder

After renaming the E2E volume, drag and drop the data from the “E2E Anonymized & Rename Drive” to the pre-created study data folder on the desktop.

To do this, drag the data to the appropriate drive shown under Output Devices (e.g., C:\). A Browse for Folder window will then appear. Select the appropriate study folder on the desktop and click "OK" to transfer the subject's data.





MANUAL	
Document Number:	MAN-018-02
Effective Date:	13-AUG-2026
Version Number:	01
Page:	26 of 35

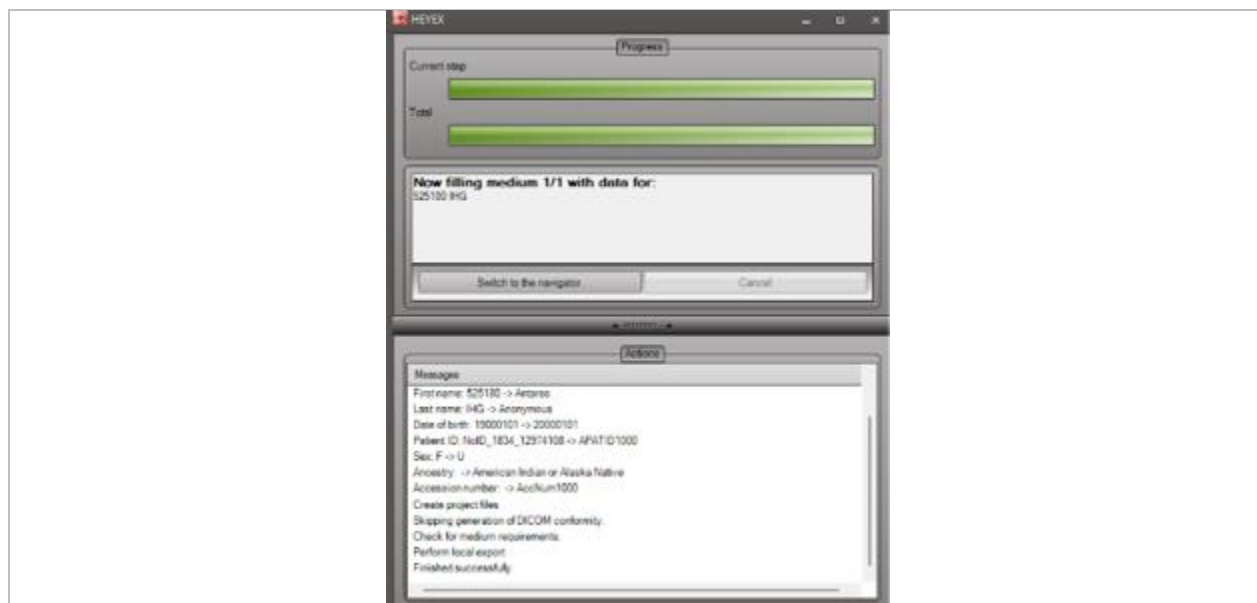
Title: Heidelberg SD-OCT Image Acquisition and File Export Manual - OCU410-301

Step 6 Confirm Data Transfer

After clicking "OK," a status bar will appear indicating the progress of the data transfer from HEYEX to the selected study folder.

Wait until the transfer is complete and verify that the data have been successfully transferred to the appropriate folder.

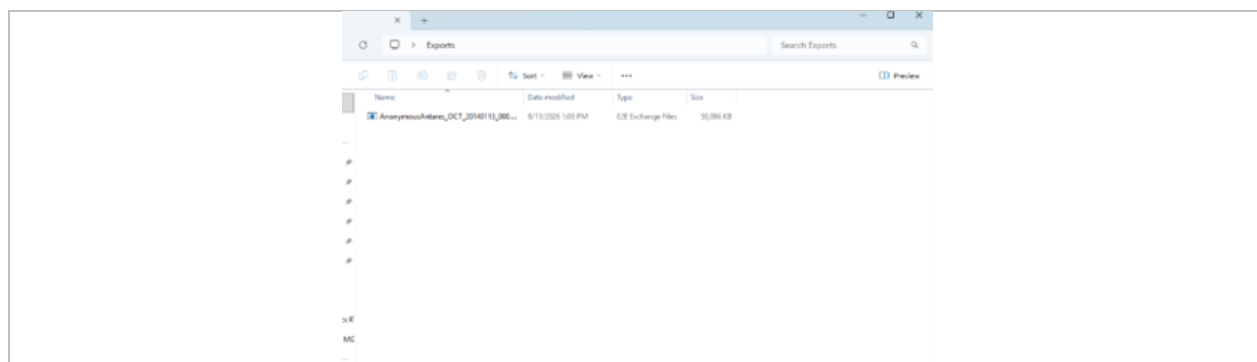
Example folder name: *SubjectID_VisitNumber_OD*



Step 7 Upload the Data to the NORC Portal

After completing the export and transfer process, upload the entire folder containing all E2E files for the corresponding subject and visit to the NORC Portal, following the study-specific upload instructions.

Important: Please ensure that all E2E files associated with the same subject/visit are included in the folder before uploading.





MANUAL	
Document Number:	MAN-018-02
Effective Date:	13-AUG-2026
Version Number:	01
Page:	27 of 35

Title: Heidelberg SD-OCT Image Acquisition and File Export Manual - OCU410-301

Questions or Issues:

If you encounter any issues exporting, renaming, or uploading OCT volume data, please contact the NORC Reading Center before proceeding.

9. NORC File Transmission Instructions

9.1 System Requirements

- A computer running a supported Windows Operating System. If your browser has issues during uploading, re-attempt using another web browser (e.g., Firefox, Chrome, etc.)
- Contact NORC before attempting to upload files, and instructions/access details will be provided.

9.2 NORC File Transfer Accounts

If you already have login credentials for the NORC secure file transfer portal, continue to use your current account. Contact NORC if you have forgotten your username or password or are unsure whether you have an account.

If you do not already have an account, contact NORC (see contact information below) regarding account creation. In the e-mail, include the following information:

- Name of the study
- Name of your principal investigator
- Name of Institution/site
- Address of Institution/site
- Site #
- Account holder's name
- Account holder's phone number
- Account holder's e-mail address

After your request is processed, you will receive email confirmation of account access and login details. If you do not receive these within a few business days, contact NORC.

9.3 Image Transmission Description

Please ensure that any digital images and digital file names are devoid of actual patient information such as name, DOB and medical record number. NORC cannot receive any protected health information (PHI) as defined in the HIPAA regulations.



MANUAL	
Document Number:	MAN-018-02
Effective Date:	13-AUG-2026
Version Number:	01
Page:	28 of 35

Title: Heidelberg SD-OCT Image Acquisition and File Export Manual - OCU410-301

9.4 Using the NORC File Transfer Portal

- i) Log in to the NORC secure file transfer portal at the URL provided by NORC.
- ii) Click the “Folders” link.
- iii) Upload the files by dragging and dropping or clicking upload files then drag and drop.
- iv) Confirm all files were uploaded as larger files may take some time.
- v) When NORC receives your upload, the images will be downloaded for review.
- vi) If you suspect a problem with the submission, or you wish to submit replacement images for any reason, contact NORC for instructions.

9.5 Submission Follow-Up

The NORC Site Portal is the primary method for submitting imaging data. A separate email notification is not required for routine submissions.

For expedited review or follow-up regarding a submitted qualification or study visit, sites may contact NORC at norc@netramind.ai or submit the request directly through the **NORC Site Portal**. Refer to the **NORC Portal User Guide** for instructions.

To assist NORC in identifying and reviewing the submission, include the following information, as applicable:

- Protocol: OCU410-301
- Site number
- Qualification or Study Visit
- Subject ID and visit, if applicable
- Submission date
- Reason for follow-up or expedited review



MANUAL	
Document Number:	MAN-018-02
Effective Date:	13-AUG-2026
Version Number:	01
Page:	29 of 35

Title: Heidelberg SD-OCT Image Acquisition and File Export Manual - OCU410-301

10. Revision History

Version Number	Date	Summary of Change(s)
01	13-AUG-2026	New Document



MANUAL	
Document Number:	MAN-018-02
Effective Date:	13-AUG-2026
Version Number:	01
Page:	30 of 35

Title: Heidelberg SD-OCT Image Acquisition and File Export Manual - OCU410-301

11. Signature Page

Author:

NAME	TITLE	SIGNATURE	DATE DD-MMM-YYYY
Sandeep Chandra Bollepalli	Director of AI	Digitally signed by Name : Sandeep Chandra Bollepalli Email : sandeep@netramind.ai Reason : I approve this document Time : Aug 13 2026 21:04 EDT Signed with : Zoho Sign 6077F16A7089178BFD3212614822B895AE...	Aug 13 2026

QA:

NAME	TITLE	SIGNATURE	DATE
Jay Chhablani	President	Digitally signed by Name : Jay Chhablani Email : chhablani@netramind.ai Reason : I approve this document Time : Aug 13 2026 22:35 EDT Signed with : Zoho Sign 6D081372BA602642A6F0C3157FF3W0243ED...	Aug 13 2026

Approver:

NAME	TITLE	SIGNATURE	DATE
Justin Shaka	Fractional COO	Digitally signed by Name : Justin Shaka Email : justin@netramind.ai Reason : I reviewed and agree to this document Time : Aug 13 2026 20:52 EDT Signed with : Zoho Sign C798418707456778BEE159BD1DFFFFED8B...	Aug 13 2026



MANUAL	
Document Number:	MAN-018-02
Effective Date:	13-AUG-2026
Version Number:	01
Page:	31 of 35

Title: Heidelberg SD-OCT Image Acquisition and File Export Manual - OCU410-301

Appendix A

OCT Image Quality: Acceptable vs. Unacceptable Images

Grading Reference for Reading Center Certified Graders

A. Purpose

Image quality is determined by the ability to assess features for each study variable. In general, the more noise or distortion present in an image, the lower the resulting image quality grade. Here are general guidelines for overall quality of the images. Details analysis of the quality will be done by our reading center and acceptance or rejection for biomarker-specific quality for the whole volume will be informed 24-48 hours after uploading the images.

Retake scans if you notice any of the following:

- A “noisy” background (numerous white dots in the aqueous space — the dark area at the top of the image — resembling snow)
- Breaks or discontinuities along the horizontal scan due to image artifact
- A non-centered fovea,
- Image tilting
- Other artifacts described below

B. Definitions and Grading Criteria

B.1 Adequate

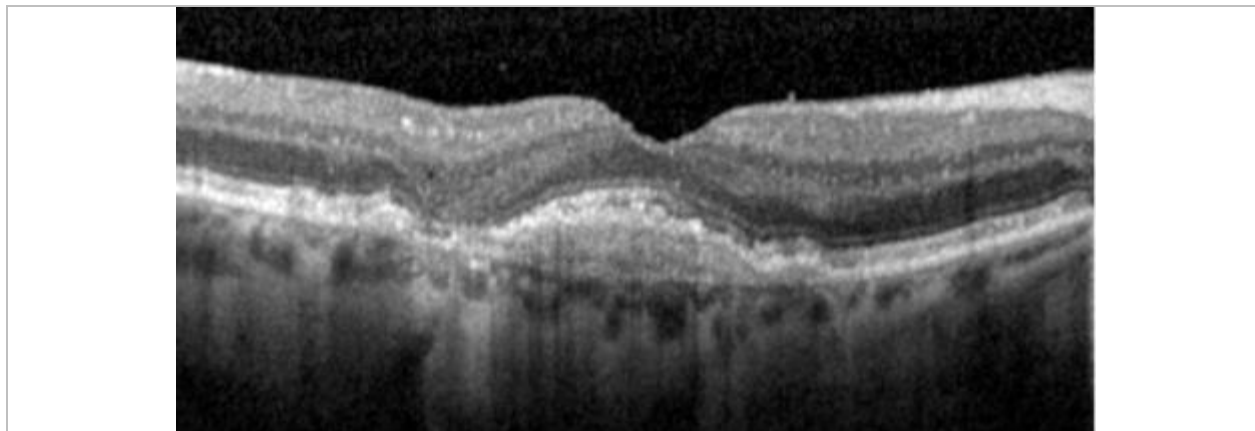
Adequate implies that photographic features can be read with great confidence.

- ☐ *Retinal layers (ILM through RPE/Bruch’s membrane) are clearly delineated across the entire scan*
- ☐ *Background (aqueous space) is dark and free of significant noise*
- ☐ *Fovea is well-centered, with the characteristic foveal depression clearly visible*
- ☐ *No significant motion artifact or misalignment between B-scans*
- ☐ *Signal strength meets or exceeds the device-specific acceptable threshold*
- ☐ *All landmarks required for study-variable grading (e.g., RPE, ELM, EZ, ONL/OPL) are distinguishable*



MANUAL	
Document Number:	MAN-018-02
Effective Date:	13-AUG-2026
Version Number:	01
Page:	32 of 35

Title: Heidelberg SD-OCT Image Acquisition and File Export Manual - OCU410-301

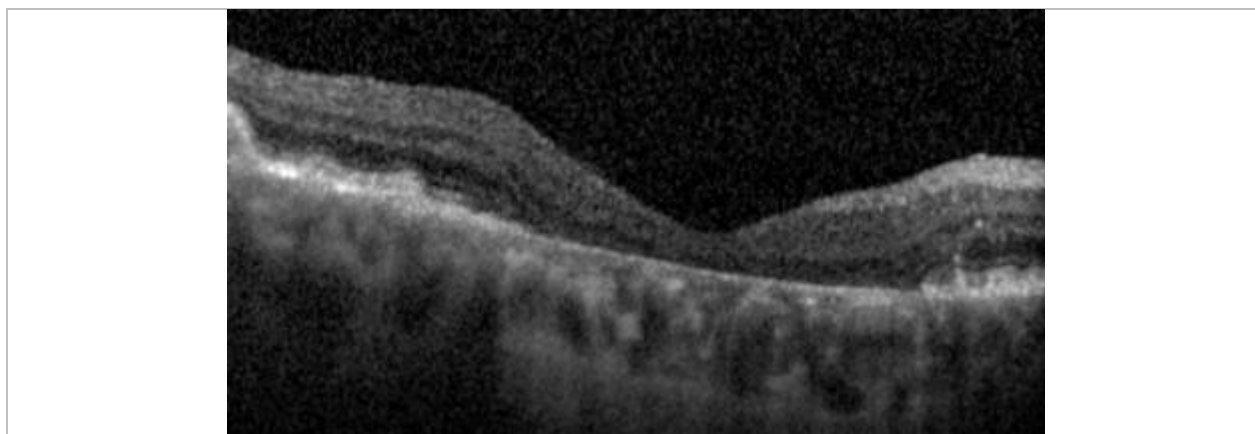


Example: Adequate image quality

B.2 Fair

Fair implies that all or nearly all photographic features can be read, but confidence in grading is only fair.

- ☐ *Most retinal layers are identifiable, though borders may be slightly indistinct in localized areas*
- ☐ *Mild-to-moderate background noise is present but does not obscure key landmarks*
- ☐ *Fovea is reasonably centered; slight tilt or decentration does not prevent grading*
- ☐ *Occasional artifact (e.g., blink line, minor motion) is present but does not affect critical grading regions*
- ☐ *Primary study variables remain assessable despite reduced grading confidence*



Example: Fair image quality

B.3 Poor

Poor implies that photographic features cannot be read with great confidence due to poor image quality; therefore, some or all features may not be gradable.

- ☐ *Significant noise or speckle obscures the boundaries of one or more retinal layers*

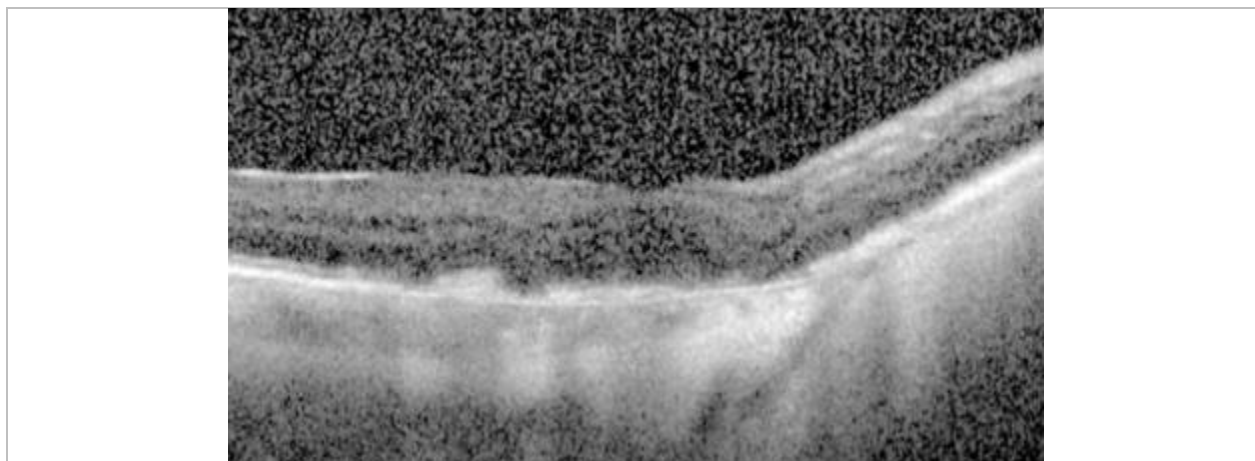
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MANUAL	
Document Number:	MAN-018-02
Effective Date:	13-AUG-2026
Version Number:	01
Page:	33 of 35

Title: Heidelberg SD-OCT Image Acquisition and File Export Manual - OCU410-301

- ☐ *Layer boundaries are blurred, discontinuous, or difficult to distinguish in critical regions*
- ☐ *Fovea is poorly centered or tilted, limiting assessment of foveal-involving features*
- ☐ *Visible motion artifact, banding, or horizontal discontinuity is present across the scan*
- ☐ *One or more study variables may not be gradable; retake is recommended if feasible*



Example: Poor image quality

B.4 Cannot Grade (CG)

Cannot Grade (CG) implies that photographic features essential for grading cannot be reliably assessed due to severe image-quality issues, artifacts, or missing image data. A scan should be classified as CG if any of the scenarios below are present:

- ☐ *Movement artifact severe enough to distort, duplicate, or discontinue the retinal contour*
- ☐ *Signal strength/image quality too low to distinguish any retinal layers*
- ☐ *Missing portion(s) of the scan or volume (cropped field, incomplete B-scans)*
- ☐ *Misalignment or missing segments within the volume scan that prevent reliable measurement*
- ☐ *No usable landmarks remain for grading any study variable*

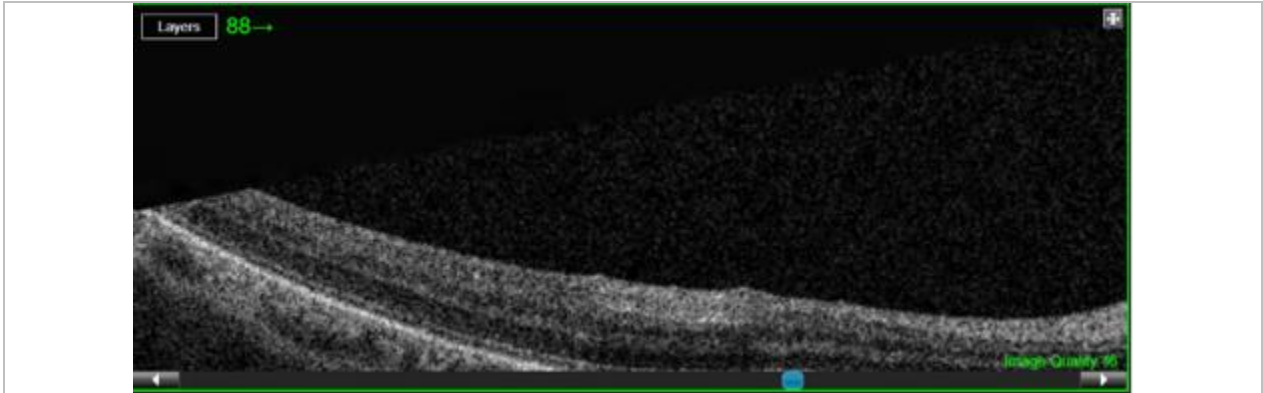
Common CG Scenarios:

OCT image with movement artifact



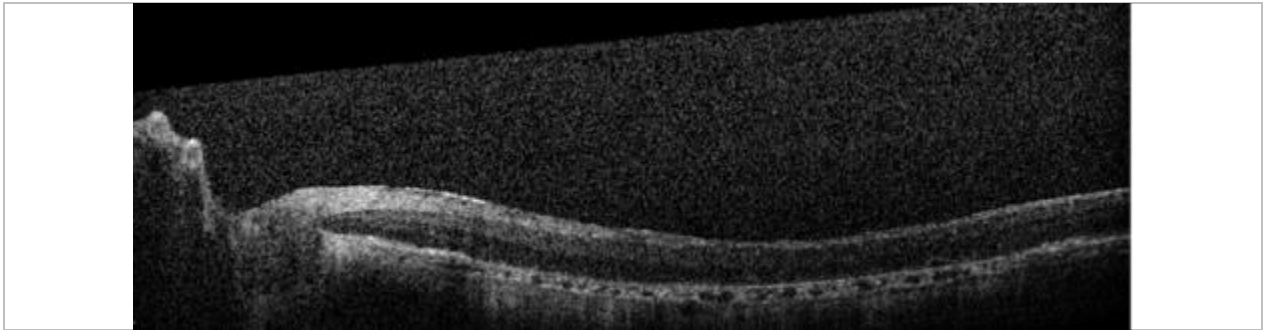
MANUAL	
Document Number:	MAN-018-02
Effective Date:	13-AUG-2026
Version Number:	01
Page:	34 of 35

Title: Heidelberg SD-OCT Image Acquisition and File Export Manual - OCU410-301



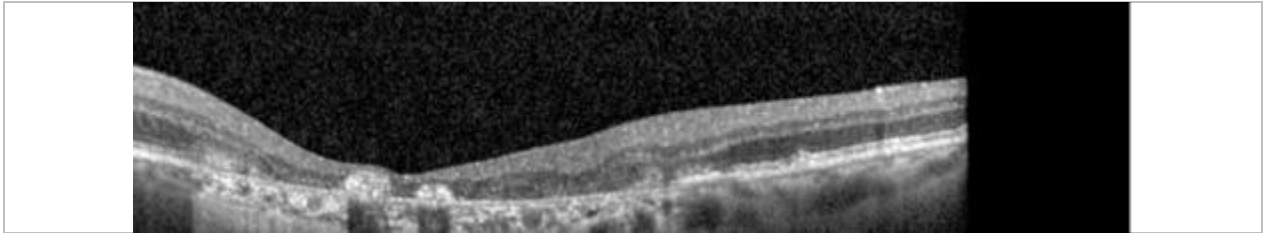
Movement artifact distorting the retinal contour

Poor image quality to grade study variables



Signal too low to reliably distinguish retinal layers

Missing part of scan/images



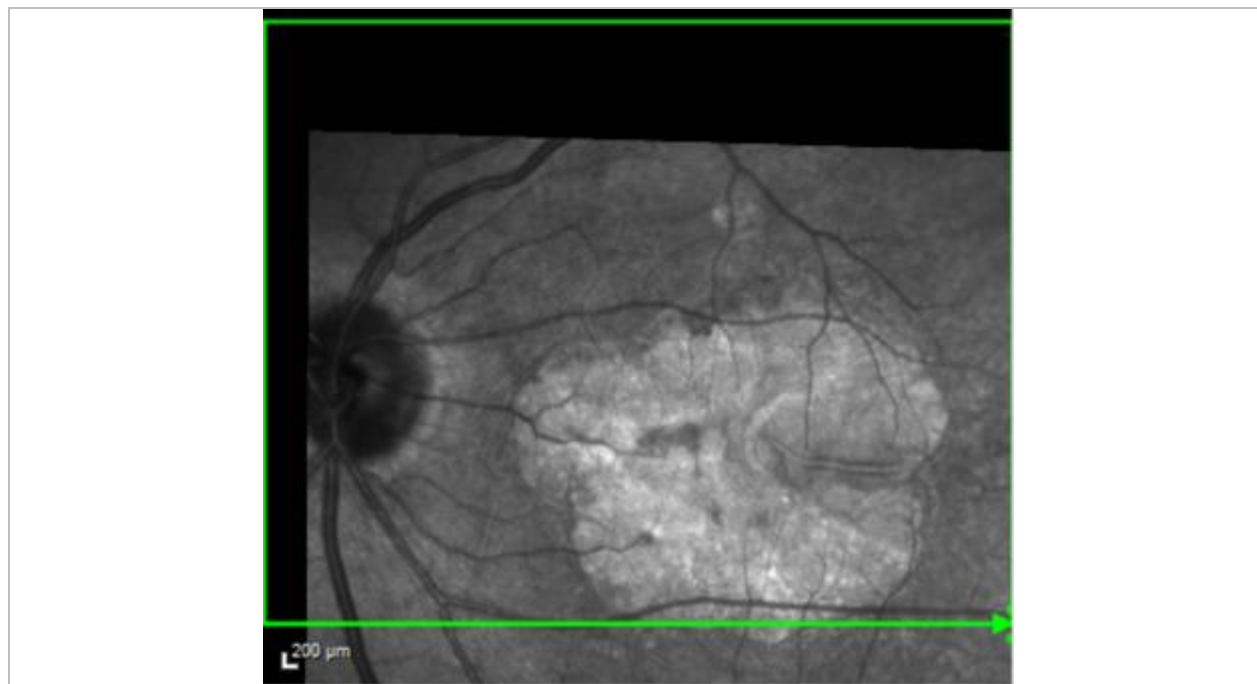
Cropped field with missing scan data

Misaligned/missing portion of the volume scans



MANUAL	
Document Number:	MAN-018-02
Effective Date:	13-AUG-2026
Version Number:	01
Page:	35 of 35

Title: Heidelberg SD-OCT Image Acquisition and File Export Manual - OCU410-301



Misalignment between fundus reference and OCT volume